

Liquid Biopsies for Cancer

A Translational Science Review

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IMPORTANCE Circulating tumor DNA (ctDNA) evaluation, in which fragments of tumor DNA circulating in a patient's bloodstream are extracted and analyzed, can be used to monitor cancer progression, detect residual cancer after treatment, and identify genetic changes within cancer cells that could affect treatment response.

OBSERVATIONS ctDNA sequencing identifies cancer cell gene variants that inform the selection of molecularly directed therapies in several types of cancer, including non-small cell lung cancer, colorectal cancer, and breast cancer. Increases or decreases in ctDNA levels can indicate treatment response (ctDNA decrease) or cancer cell resistance and recurrence (ctDNA increase). Detecting ctDNA after curative intent therapy correlates strongly with cancer recurrence and poorer survival. In a meta-analysis of 1725 patients undergoing treatment for urothelial carcinoma, higher ctDNA levels were associated with poorer survival outcomes (hazard ratio for disease-free survival, 20.69 [95% CI, 9.63-44.43]; $P < .001$). This association was also observed in adjuvant settings (hazard ratio for disease-free survival, 4.51 [95% CI, 3.04-6.69]; $P < .001$) and in patients undergoing systemic therapy for metastatic disease (hazard ratio for overall survival, 2.0 [95% CI, 1.25-3.38]; $P = .004$; absolute rates not available). ctDNA detection may indicate minimal residual disease, defined as cancer cells detectable only by highly sensitive testing (eg, detection of 1 cancer cell in a population of 1 million normal cells) before disease progression is identified with imaging. Detecting an early increase in ctDNA and/or a novel sequence variation that may confer resistance to standard treatment can guide therapeutic decisions, such as changing to a new treatment, before tumor progression is detectable with conventional imaging. In a prospective cohort study of 130 patients with colorectal cancer, molecular relapse of disease was detected approximately 8.7 months earlier compared with standard-of-care imaging surveillance (5.5 months vs 14.2 months; $P < .001$). Similarly, patients with undetectable ctDNA levels may be able to discontinue therapy and be monitored, preventing potentially unnecessary exposure to chemotherapy that may have substantial adverse effects. However, the optimal timing of ctDNA testing, management of positive results in the absence of radiographic disease, and the cost-effectiveness of serial monitoring remain unclear.

CONCLUSIONS AND RELEVANCE ctDNA, consisting of small DNA fragments from cancer cells that can be analyzed in human blood, can help clinicians monitor cancer progression, detect minimal residual cancer, and identify genetic variants that may help guide treatment decisions. Use of ctDNA may help select best treatment and timing of therapy for a patient with cancer, but optimal clinical applications remain unclear.

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A liquid biopsy is defined as using bodily fluids, such as urine, cerebrospinal fluid (CSF), blood, saliva, pleural fluid, peritoneal fluid, or stool, to detect markers of malignancy, infection, cardiovascular disease, and autoimmune disorders through analysis of cell-free DNA (cfDNA), RNA, proteins, exosomes, and microRNAs (Box 1). cfDNA was first described in 1948, when extracellular DNA was found circulating in human blood.¹ In healthy individuals, cfDNA has distinct proportional contributions from various cell types, including hematopoietic cells, vascular endothelial cells, and hepatocytes.² Changes in the proportion of cfDNA from these tissues or detection of specific gene variants (eg, an elevated proportion of cells with the *BRCA* variant) can indicate underlying pathological processes. In transplant medicine, organ recipients can be monitored for rejection or tissue injury via donor-derived cfDNA, for whom a specific threshold of donor-derived cfDNA from plasma correlates with rejection or injury; in infectious disease, pathogen-derived cfDNA can be detected in the blood of a patient with sepsis to identify pathogenic organisms and monitor antibiotic response; and in fetal medicine, fetal-derived cfDNA in maternal blood can be used to detect aneuploidy.³⁻⁵

In oncology, liquid biopsies typically refer to blood samples, specifically the plasma fraction collected into either a Streck Cell-Free DNA Blood Collection Tube or a K/K3-EDTA tube, used to extract DNA, RNA, or proteins. However, other bodily fluids, such as CSF in people with primary brain tumors or new metastatic disease with an unknown primary tumor, can also be used. These tests are currently commercially available. In oncology, a liquid biopsy may include sequencing DNA or RNA for malignancy-related sequence variations and may also include liquid chromatography and mass spectrometry to identify levels of protein variations, such as epidermal growth factor receptor (*EGFR*), *BRCA*, or protein modifications, including lysine acetylation and arginine mono-methylation from blood, CSF (brain tumor[s]), or urine (kidney or urothelial malignant neoplasms).⁶ Liquid biopsies can provide information without obtaining a tissue (tumor) sample, which can be helpful for malignancy of unknown primary, tumors inaccessible to biopsy, and in patients for whom a biopsy procedure is associated with substantial risk, such as proximity to a vital organ (eg, heart, aorta).³ For example, cfDNA, consisting of very low levels of circulating fragmented tumor DNA, can identify specific, pathologic, and clinically relevant *EGFR* variants in DNA shed from non-small cell lung cancer (NSCLC) cells.⁷

The mechanisms by which cellular DNA enters the circulation are not entirely clear. However, analysis of DNA fragments, or pieces of DNA that are approximately 150 to 200 base pairs (the amount of DNA wrapped around a nucleosome, which is the fundamental packing unit of DNA),⁴ suggests that some cfDNA originates from specific cleavage of DNA in cells that are undergoing apoptosis or apoptotic cell death, defined as cells undergoing controlled, programmed cell death.⁸ cfDNA is also actively shed from nonapoptotic cells.^{9,10} Tumor cells also shed DNA, referred to as *circulating tumor DNA* (ctDNA), consisting of fragments of DNA that can often be shorter than cfDNA from normal cells. These shorter DNA fragments occur because tumor cells are more likely to undergo more uncontrolled mechanisms of cell death, such as necrosis, a process by which cells die in an uncontrolled, premature fashion. Compared with enzymes that cleave DNA during apoptosis, those that cleave DNA during necrosis are much less organized, resulting in

Box 1. Frequently Asked Questions About Circulating Tumor DNA (ctDNA)

What is a liquid biopsy?

A liquid biopsy consists of blood or other bodily fluid, such as cerebrospinal fluid or urine, that is collected to detect cancer markers or to identify tumor gene variants that can guide therapy. Examples of liquid biopsies include ctDNA, circulating tumor cells, exosomes, and microRNAs.

What is ctDNA?

ctDNA are small fragments of DNA released by tumor cells into the bloodstream. ctDNA can enter the blood passively as tumor cells die or through active secretion of tumor DNA by cancer cells. Approximately 75% of patients with cancer have detectable ctDNA.

How can ctDNA be used to improve outcomes in patients with cancer?

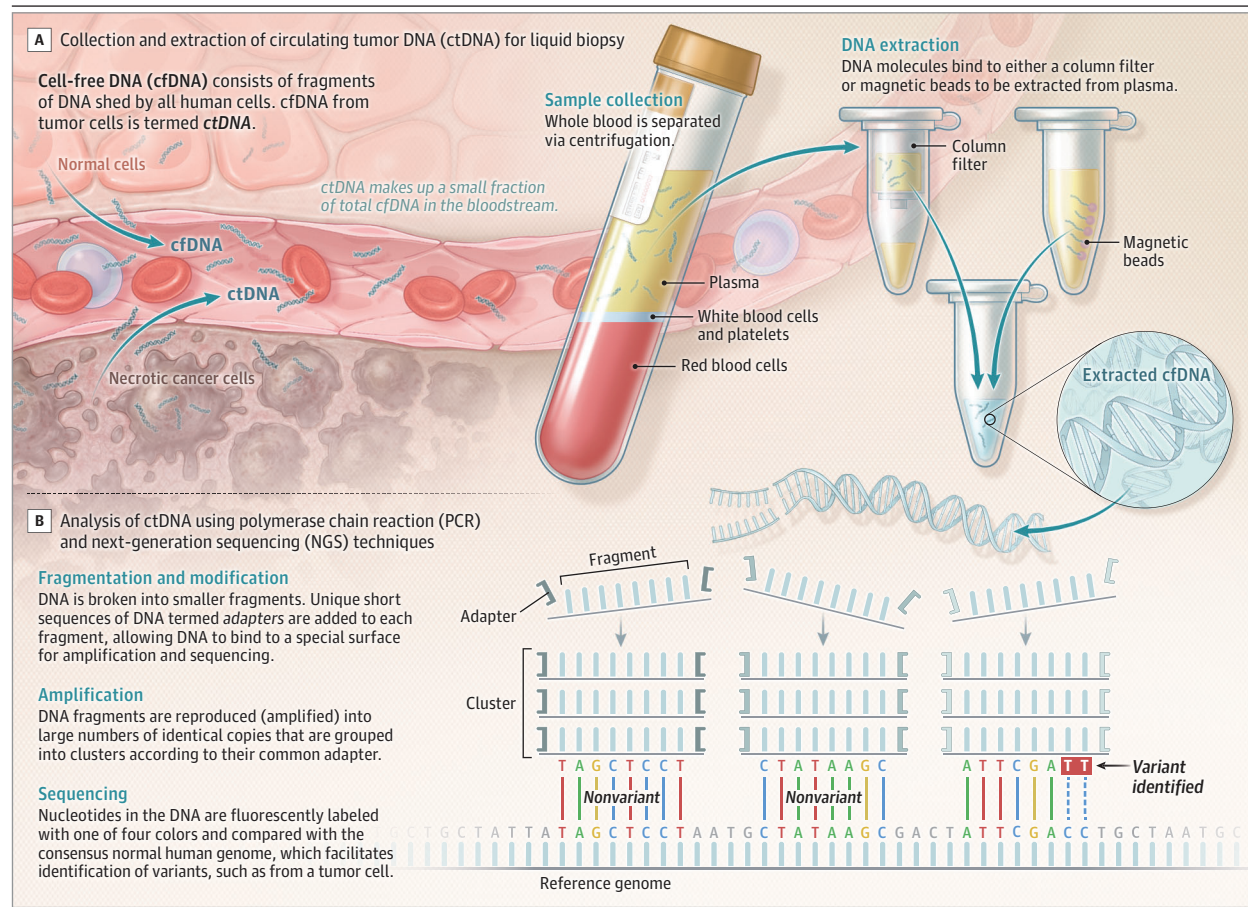
ctDNA can improve selection of personalized therapy and detect minimal or molecular residual disease after curative intent treatment, which provides information about presence of residual cancer that can be prognostic. ctDNA can also be used to monitor treatment response over shorter time periods compared with monitoring with imaging and detect emergence of variants that can confer treatment resistance.

a heterogeneous population of DNA fragments. Shorter DNA fragments from tumor cells may also be due to differences in chromatin arrangement, increased enzymatic activity, and epigenetic alterations that change DNA accessibility to cleaving enzymes.^{4,11,12} In patients with a malignant neoplasm, ctDNA is typically only a small fraction of cfDNA (cfDNA includes extracellular [ie, cell-free] DNA from any cell in the body, normal or tumor; ctDNA includes only DNA fragments from tumor cells), often less than 1%, but this proportion depends on tumor characteristics such as stage, grade, presence of metastasis, vascularization of the tumor, and rate of cell turnover.^{13,14} Higher grade, higher stage, metastatic disease, and increased vascularization are associated with higher ctDNA concentrations. ctDNA is detectable within cfDNA and is often reported as variant allele fractions (VAFs), which represent the percentage of variant DNA molecules relative to the total number of wild-type DNA molecules for a particular gene, although both tumor cells and non-tumor cells shed non-variant-bearing DNA (nonvariant cfDNA can be distinguished between tumor and nontumor cells through targeted next-generation sequencing).¹⁵⁻¹⁷ Higher VAFs are associated with greater tumor burden and poorer prognosis in multiple cancer types, including colorectal cancer (CRC), NSCLC, pancreatic cancer, and breast cancer.¹⁸⁻²³ Therapeutic interventions, such as chemotherapy, immunotherapy, tumor-informed therapy (ie, CRC with the *BRAF V600E* variant treated with combination therapy vs a single agent), or surgery are associated with a reduction in ctDNA.²⁴⁻²⁸ Declining levels of ctDNA after intervention are associated with greater treatment response.²⁸

ctDNA Collection and Analysis

ctDNA is detectable in serum, urine, and CSF. However, ctDNA from plasma is the preferred sample compared with serum because serum

Figure. Illustration of Collection, Extraction, and Analysis of Circulating Tumor DNA (ctDNA)



typically contains more genomic DNA released from lysis of white blood cells during the clotting process, which dilutes the tumor-derived fraction of DNA. Measuring ctDNA requires separating plasma from whole blood using double centrifugation or rotating a sample at high speed to separate blood components based on densities, followed by removal of cellular components of blood (red blood cells, white blood cells, platelets) and subsequent DNA extraction using a molecular fishing model, in which particles of DNA attach themselves to either a column or magnetic bead, while other material passes through as waste. The time by which these samples need to be processed differs based on the collection tube. For standard EDTA tubes, processing should begin within 2 to 4 hours; tubes designed for ctDNA collection can have delayed processing of up to 7 days. These methods help minimize sample contamination with high-molecular-weight DNA that is released from the rupture of white blood cells during sample collection and preparation, therefore facilitating variant measurement at VAFs of 0.01% or lower.²⁸⁻³⁰ The ability to detect a VAF of 0.01% means that 1 variant haploid genome can be detected from 10 000 normal haploid genomes. Sensitivity and specificity vary between different types of ctDNA assays. The 2 most common methods for analyzing ctDNA are polymerase chain reaction (PCR) and next-generation sequencing (NGS) techniques. PCR consists of rapidly reproducing small DNA fragments into large numbers of copies, which allows identification of loci of interest and aids in detecting rare ctDNA from among all cfDNA

collected. NGS is a high-throughput platform that involves further fragmentation and modification of DNA (or RNA), which allows it to bind to a specialized surface. These fragments are then amplified by PCR (described above) to create clusters of identical copies on the flow cell. The machine sequentially adds fluorescently labeled nucleotides (adenine, cytosine, guanine, thymine) of different colors. When incorporated into the growing strand, the resulting colors emitted display the DNA code (sequence or arrangement of color-coded adenine, cytosine, thymine, and guanine that make up a fragment) in millions of DNA fragments simultaneously (Figure).

Clinical Applications of ctDNA Results

ctDNA allows detection of gene variants that are pathognomonic for specific cancer types and that are associated with tumor growth and/or resistance to treatment based on existing gene variants. Because ctDNA values directly correlate with extent of disease, quantitative ctDNA values can be useful for monitoring response to treatment or disease recurrence.

Gene Variants Associated With Tumor Growth

Identifying gene variants to guide cancer treatments is now standard practice for many types of cancer, including NSCLC, breast cancer, and CRC. Previously, NGS of solid tumor specimens was used

to identify biologically and clinically important gene variants that offered prognostic information about treatment response, resistance to treatment, and survival.³¹ For example, *ESR1* gene variants in breast cancer are associated with acquired resistance to aromatase inhibitors,³²⁻³⁴ use of vorasidenib is indicated in patients with low-grade glioma who have *IDH1* or *IDH2* genetic variants,³⁵ and *KRAS* sequence variation status in colorectal cancer signifies *EGFR* resistance and helps identify which patients may benefit from sotorasib and panitumumab.³⁶ Identifying these variants facilitates selecting personalized, targeted therapies for patients.

Initiating anticancer therapy with medications that directly target a gene variant typically leads to superior outcomes for patients.³⁷ For example, the CodeBreak 100 study was a single-group, nonrandomized phase 2 trial involving 126 patients with previously treated NSCLC who had detectable p.G12C plasma gene variants in the *KRAS* oncogene (Table).³⁸⁻⁴⁴ Eligible patients were treated with sotorasib, a *KRAS* G12C inhibitor, and the primary end point was objective response assessed by blinded, independent, central radiologic review. Of 126 enrolled patients, 124 were evaluated for response following sotorasib treatment, of whom 46 patients had a partial or complete response (37.1% [95% CI, 28.6%-46.2%]). Disease control, defined as a complete response, partial response, or stable disease, occurred in 100 of 124 patients (80.6% [95% CI, 72.6%-87.2%]).³⁸ In this study, the percent positive agreement between *KRAS* p.G12C plasma gene variants and the tissue-informed assay (*KRAS* RGQ PCR kit) was 0.71 (95% CI, 0.62-0.79), while the negative percent agreement was 1.00 (95% CI, 0.95-1.00).⁴⁵

The results from the phase 2 CodeBreak 100 study showed that patients with NSCLC with p.G12C *KRAS* gene variants may benefit from sotorasib treatment regardless of how gene variants are identified (tumor-informed [tissue biopsy-based] assays vs plasma ctDNA analysis),⁴⁵ demonstrating 2 potential advantages of plasma ctDNA analysis. First, plasma ctDNA analysis measures ctDNA shed from multiple sites within a tumor and from tumors in multiple locations simultaneously, thereby providing useful information when biopsy samples have insufficient size for NGS analysis or when a biopsy sample is not available or not possible. However, a negative test result for ctDNA does not exclude the possibility that a genetic variant is present in cancer tissue. The European Society for Medical Oncology Precision Medicine Working Group recommends reflexively obtaining tumor tissue for genetic analysis when the fraction of ctDNA in blood samples is below the detectable limit or when analysis does not reveal clinically relevant genetic variants.³⁷ Additionally, comparison of genetic variation in tumor tissue with ctDNA can improve time to result and detection of clinically relevant genetic variants.⁴⁶ For example, a single-center retrospective analysis in 166 patients with lung adenocarcinoma showed that the initiation of standard reflex tissue ctDNA testing at the time of initial pathologic diagnosis decreased time to initiation of treatment by 37 days compared with the previous standard, in which testing was ordered at the time of the patient's first oncology appointment (52.6 days to 15.6 days; $P = .0002$).⁴⁶

Detecting Gene Variants Associated With Treatment Resistance

ctDNA detects changes in tumor genomics at predetermined time points or in real time throughout a treatment course to monitor

development of resistance to treatment. In the SERENA-6 randomized clinical trial, 3256 patients with estrogen receptor (ER)-positive, *ERBB2* (formerly *HER2* or *HER2/neu*)-negative advanced breast cancer who had received at least 6 months of treatment with an aromatase inhibitor plus a cyclin-dependent kinase (CDK) 4/6 inhibitor for first-line treatment of metastatic disease underwent serial ctDNA testing using the Guardant360 CDx assay to evaluate for emergence of *ESR1* variants, which confer resistance to standard endocrine therapies, such as aromatase inhibitors (Table).³⁹ Patients ($n = 315$) with positive test results for *ESR1* variants without evidence of disease progression were randomized 1:1 to either continue their aromatase inhibitor and CDK4/6 inhibitor therapy (palbociclib, ribociclib, or abemaciclib) or to switch therapy to camizestrant and a CDK4/6 inhibitor.³⁹ Camizestrant is an ER- α selective treatment that degrades ER and acts as a complete ER antagonist. This treatment has activity against both *ESR1* variants and wild-type ER.⁴⁷ Of 3325 screened patients, 548 (16%) had a positive test result for an *ESR1* variant, and 315 were randomized.³⁹ The primary outcome was investigator-assessed progression-free survival. At a median follow-up of 12.6 months, compared with patients who received an aromatase inhibitor combined with CDK4/6 inhibitor, those who received camizestrant combined with the CDK4/6 inhibitor had significantly improved progression-free survival (9.2 vs 16 months; hazard ratio [HR] for progression or death, 0.44 [95% CI, 0.31-0.60]; $P < .001$).³⁹ Whether this treatment strategy affects overall survival (OS) is unknown.³⁹

The utility of frequent ctDNA testing and optimal frequency of ctDNA measurement during cancer treatment is unknown. The SERENA-6 study collected and analyzed ctDNA samples for *ESR1* variants approximately every 2 to 3 months in patients being treated with standard of care, a first-line CDK4/6 inhibitor and an aromatase inhibitor.³⁹ However, this approach is expensive, may not be associated with improved mortality, and is not currently standard of care for patients with breast cancer.

Detecting Residual Cancer After Treatment

The ability of the presence or absence of ctDNA to provide prognostic information following neoadjuvant or adjuvant cancer treatment has been evaluated in CRC, for which ctDNA is a minimally invasive and reliable method for detecting molecular residual disease, also referred to as *minimal residual disease*. After colon cancer resection, detectable levels of ctDNA are associated with higher risk of recurrence.⁴⁸

An observational study measured ctDNA-based minimal residual disease status in patients with stages II to IV recurrent colorectal cancer who underwent surgical resection of cancer (Table).⁴⁰ A ctDNA monitoring, tumor-informed assay evaluated 16 somatic single-nucleotide variants present in plasma based on variants identified via whole-exome sequencing of tumor tissue.⁴⁰ Among 1039 enrolled patients, positive ctDNA testing results (ie, ctDNA present 4 weeks after surgery) were associated with a higher rate of recurrence compared with patients with negative ctDNA testing results (61.4% vs 9.5%, respectively; $P < .001$).⁴⁸ Among the 113 patients with a positive ctDNA result at 4 weeks, those who received adjuvant chemotherapy had an 18-month disease-free survival (DFS) rate of 61.6% vs 22.0% ($P < .001$) in those who did not receive adjuvant chemotherapy.⁴⁸ Among the

Table. Clinical Trials Using Circulating Tumor DNA (ctDNA) and Their Outcomes

Clinical trial	Cancer type	Eligible patients	Study/hypothesis	Sample size	ctDNA test type	Intervention	Results (primary end point)	Other considerations
CodeBreak100 (phase 2, nonrandomized, single group) ³⁸	Advanced NSCLC	Patients with advanced, pretreated NSCLC with confirmed p.G12C KRAS gene variant	Sotorasib would yield a clinically meaningful and durable objective response rate in patients with NSCLC with a KRAS p.G12C gene variant	126	Targeted NGS ^a	Sotorasib (single-group study)	Objective response rate (37.1%) measured by RECIST 1.1 criteria via computed tomography or magnetic resonance imaging	80.6% Disease control rate
SERENA-6 (phase 3, randomized controlled) ³⁹	Metastatic ER-positive/ERBB2-negative breast cancer	Patients with ER-positive/ERBB2-negative advanced breast cancer who received at least 6 mo of treatment with an aromatase inhibitor plus CDK4/6 inhibitor for first-line treatment of metastatic disease	Using liquid biopsies to detect ESR1 variants before radiographic progression and switching aromatase inhibitor to camizestrant would improve progression-free survival	3256	Targeted NGS	If ESR1 variant developed without evidence of radiologic disease progression, randomized to camizestrant (plus CDK4/6 inhibitor) group or control group (continue aromatase inhibitor plus CDK4/6 inhibitor)	Progression-free survival: 16.0 mo in the camizestrant plus CDK4/6 inhibitor group and 9.2 mo in the aromatase inhibitor plus CDK4/6 inhibitor group	Serial ctDNA monitoring for ESR1 variants was performed every 2-3 mo
CIRCULATE-Japan GALAXY (prospective observational study) ^{40,41}	Stage II-IV or relapsed colorectal cancer	Patients with stage II-IV or relapsed colorectal cancer who had completed surgical resection of disease with no radiographic evidence of disease recurrence	Monitor ctDNA-based minimal residual disease status and correlate with recurrence risk	2240	16-plex PCR NGS	ctDNA monitoring was used for observational correlation with outcomes, but did not mandate treatment changes (not an interventional trial)	ctDNA positivity at 4 wk postoperatively was associated with a 10- to 12-fold increased risk of recurrence across all stages	Patients with negative minimal residual disease status had similar disease-free survival rates at 18 mo regardless of whether they received chemotherapy or not
DYNAMIC (randomized noninferiority study) ^{42,43}	Stage II colon or rectal adenocarcinoma	Patients with stage II colon or rectal adenocarcinoma who underwent surgical removal of cancer and had negative margins and no lymph node involvement	A ctDNA-guided approach to adjuvant treatment would reduce the use of adjuvant treatment without compromising recurrence risk	455	Tumor-informed NGS ^b	ctDNA-informed therapy (chemotherapy if detectable minimal residual disease) vs standard-of-care management (adjuvant therapy decisions made based on clinicopathological risk features)	Recurrence-free survival at 2 y was noninferior: 93.5% and 92.4% for ctDNA-guided therapy compared with standard of care	Underpowered for subgroup analysis; delayed chemotherapy initiation due to testing turnaround time (11.9 wk vs ≤8 wk, which is guideline directed)
IMvigor011 (phase 3 randomized controlled) ⁴⁴	Muscle-invasive bladder cancer	Patients with muscle-invasive bladder cancer who were determined to be cancer free at 6-24 wk after surgery	ctDNA-directed therapy with adjuvant atezolizumab would lead to better outcomes than placebo	761	Tumor-informed NGS	Atezolizumab every 4 wk for 12 cycles or up to 1 y vs placebo in patients who were found to have positive results for minimal residual disease by ctDNA testing	Disease-free survival: 9.9 mo in atezolizumab group, 4.8 mo in placebo group	Median overall survival: 32.8 mo in atezolizumab group, 21.1 mo in placebo group

Abbreviations: CDK, cyclin-dependent kinase; DYNAMIC, Circulating Tumour DNA Analysis Informing Adjuvant Chemotherapy in Stage II Colon Cancer; ER, estrogen receptor; GALAXY, Genetic Alterations and Clinical Record in Radically Resected Colorectal Cancer Revealed by Liquid Biopsy and Whole Exome Analysis; NGS, next-generation sequencing; NSCLC, non-small cell lung cancer; PCR, polymerase chain reaction; RECIST, Response Evaluation Criteria in Solid Tumors.

^a Focuses on specific genes or regions of interest to sequence rather than the entire genome.

^b Uses a patient's tissue-based biopsy to create a personalized, individual-specific sequencing assay.

531 participants with a negative ctDNA test result, 18-month DFS was similar between patients who underwent adjuvant chemotherapy vs those who were observed without treatment (94.9% and 91.5%, respectively; $P = .16$).⁴⁸

Another cohort of 2240 patients with stages II to III colon cancer or stage IV CRC underwent ctDNA testing and were followed up for a median of 23 months.⁴¹ Patients whose results were persistently negative for minimal residual disease by ctDNA testing at 3 and 6 months after adjuvant chemotherapy had significantly improved DFS and OS at 24 months compared with patients with minimal residual disease that became detectable (89% DFS vs 3.3% [$P < .001$] and 100% OS vs 82.3% [$P = .007$]).⁴¹ Detectable minimal residual disease results preceded recurrence detected by conventional methods, such as computed tomography imaging, or diagnostic procedures, such as colonoscopy, by a median of 5.91 months (range, 0-33.15 months).⁴¹ These results were similar to those of a prospective cohort study of 130 patients with CRC, where molecular relapse of disease was detected 8.7 months earlier using ctDNA testing than with standard-of-care surveillance imaging (5.5 months vs 14.2 months; $P < .001$).⁴⁹ These findings suggest that molecular recurrence of disease, measured by ctDNA, may be a more sensitive measure of recurrence than conventional imaging. However, studies are needed to assess whether treatment initiation at the time of minimal residual disease detection via ctDNA testing improves outcomes.

The Circulating Tumour DNA Analysis Informing Adjuvant Chemotherapy in Stage II Colon Cancer (DYNAMIC) study was a noninferiority clinical trial that randomized patients with stage II colon or rectal adenocarcinoma who underwent surgical resection of cancer and had negative resection margins and no lymph node involvement (Table). DYNAMIC tested whether prescribing adjuvant chemotherapy based on ctDNA minimal residual disease testing (in which patients with positive minimal residual disease results received adjuvant chemotherapy and those with negative minimal residual disease results were observed without therapy) was noninferior to standard-of-care treatment, which consisted of making decisions on proceeding with adjuvant chemotherapy vs observation based on conventional clinicopathologic criteria. A total of 455 eligible participants were randomized 2:1 to undergo either treatment based on ctDNA minimal residual disease results vs standard-of-care management. The primary efficacy end point was recurrence-free survival at 2 years.⁴² The prespecified noninferiority margin was a -8.5% difference.⁴² Plasma was analyzed for ctDNA at weeks 4 and 7 after surgery, and patients in the ctDNA therapy-directed cohort who had positive ctDNA results at either of those time points received adjuvant single-agent fluoropyrimidine or oxaliplatin-based chemotherapy, while patients with negative ctDNA results did not receive adjuvant chemotherapy.⁴²

ctDNA analysis was successfully performed for 291 of 294 patients (99%) in the ctDNA-guided therapy group.⁴² Of these, 45 had detectable minimal residual disease by ctDNA testing and 44 of these patients received chemotherapy. One patient whose test results were negative for minimal residual disease received chemotherapy.⁴² In the standard management group, 41 of 147 patients (28%) received chemotherapy. At 2-year follow-up, the ctDNA-guided treatment was noninferior to standard-of-care

management for the outcome of recurrence-free survival (93.5% for ctDNA-guided management and 92.4% for standard management; absolute difference, 1.1% [95% CI, -4.1% to 6.2%]; noninferiority margin, -8.5%). At 5 years of follow-up, the prespecified secondary analysis of OS was 93.8% in the ctDNA-guided group and 93.3% in the standard treatment group (HR, 1.05 [95% CI, 0.47-2.37]; $P = .89$), consistent with the management strategy in which patients could forgo treatment with chemotherapy in the ctDNA-guided treatment group and achieve similar survival outcomes.⁴³ Additionally, the probability of remaining recurrence free at 5 years was 0% for patients with detectable minimal residual disease at the end of adjuvant chemotherapy ($n = 6$) compared with 96.8% for patients without detectable ctDNA at the end of adjuvant chemotherapy ($n = 32$; $P < .001$), demonstrating the potential prognostic value of ctDNA.⁴³

In the phase 3, double-blind, randomized IMvigor011 trial of patients with muscle-invasive bladder cancer who were free of cancer at 6 to 24 weeks after surgery, 250 of 761 enrolled patients (33%) had a positive test result for minimal residual disease by ctDNA analysis and were randomized 2:1 to receive adjuvant atezolizumab or placebo every 4 weeks for 12 cycles or up to 1 year (Table).⁴⁴ ctDNA testing was performed every 6 weeks for a total of 9 months, and an additional ctDNA test was performed 1 year after surgery.⁴⁴ Patients whose test results were negative for ctDNA ($n = 357$) did not receive atezolizumab or placebo.⁴⁴ At a median follow-up of 16.1 months, DFS (primary end point) occurred at 9.9 months in the atezolizumab group vs 4.8 months in the placebo group (HR, 0.64 [95% CI, 0.47-0.87]; $P = .005$). The prespecified secondary end point of median OS was 32.8 months in the atezolizumab group vs 21.1 months in the placebo group (HR, 0.59 [95% CI, 0.39-0.90]; $P = .01$).⁴⁴

Current US Food and Drug Administration-Approved Uses of Liquid Biopsy in Cancer

Although there are many commercially available ctDNA tests for various indications, US Food and Drug Administration (FDA)-approved ctDNA testing modalities are more limited, which is important for clinicians to consider when discussing ctDNA testing with patients. In 2016, the FDA approved a diagnostic liquid biopsy for cancer (the cobas EGFR Mutation Test v2), which detects specific *EGFR* gene variants amenable to targeted therapy with *EGFR* inhibitors in NSCLC.⁷ Two other tests, Guardant360 CDx and FoundationOne Liquid CDx, are also approved for gene variant detection for patients with NSCLC. The FDA approval allows clinicians to select treatment based on the presence of specific gene variants.^{45,50} In breast cancer, the Guardant360 CDx NGS panel is approved for detecting *ESR1* and *PIK3CA* variants in patients with metastatic ER-positive/*ERBB2*-negative disease.^{51,52} FoundationOne CDx can also be used to identify gene variants in patients with breast cancer for which specific therapies can improve outcomes.^{33,50} The Therascreen *PIK3CA* RGQ PCR kit is also FDA approved to identify patients with ER-positive/*ERBB2*-negative metastatic breast cancer with the *PIK3CA* variant that make them eligible for treatment with alpelisib, a *PI3K* inhibitor.³⁴ In May 2026, the FDA approved the Signatera CDx test as the standard diagnostic test (also known as

Box 2. Advantages and Disadvantages of Circulating Tumor DNA (ctDNA) Testing**Advantages of ctDNA Testing**

- Minimally invasive
- Assesses heterogeneous populations by testing DNA from multiple sites
- Results are available in approximately 7 d compared with ≥ 14 d for tissue-based sequencing
- Facilitates evaluation of tumor resistance variants without the need for an additional tissue biopsy
- Ability to assess for minimal residual disease in patients where radiographical evaluation may not be sensitive enough to detect disease
- Minimal residual disease testing provides prognostic information

Disadvantages of ctDNA testing

- Potential for false-negative results for tumors that are small, early stage, or low shedding
- Does not provide information about tumor microenvironment, the area that surrounds and supports the tumor; the tumor microenvironment contains immune cells, blood vessels, and additional stroma, which are the target of other therapies, such as immune checkpoint inhibitors and antiangiogenesis medications
- ctDNA may be more expensive compared with traditional imaging for diagnosis or surveillance of malignancy
- Limited availability in resource-limited settings
- Lack of US Food and Drug Administration approval for several commercially available tests

the companion diagnostic device) to identify patients with muscle-invasive bladder cancer with minimal residual disease after cystectomy who are candidates for treatment with adjuvant atezolizumab, a monoclonal antibody that targets PD-L1.⁴⁴

Challenges Associated With Clinical Applications of ctDNA

There are several challenges associated with use of ctDNA for evaluation of patients with cancer (Box 2). First, variability exists in the quantity of ctDNA secretion among different cancer types. In a study that used distinct assays to detect ctDNA, it was detectable in more than 50% of patients with bladder, colorectal, gastroesophageal, ovarian, and pancreatic ductal cancers, but in fewer than 10% of patients with gliomas.⁵³ It is unknown why some cancers secrete relatively more or less DNA into the circulation than others. Second, for patients with metastatic cancer, the optimal frequency of ctDNA testing is unclear. Third, it is unclear if ctDNA levels should be used as a marker of early response to treatment that is more sensitive than imaging techniques. Fourth, in patients treated with definitive surgery, radiation, and/or chemotherapy or other cancer therapies, the optimal frequency of ctDNA testing to identify recurrence is unclear. Fifth, in patients who have completed definitive cancer therapies, it is unclear whether treatment should be started at the time of minimal residual disease detection if metastatic disease is not identified on imaging. Sixth, cfDNA/ctDNA as a primary cancer screen cur-

rently lacks high accuracy, with early studies demonstrating that multicancer early detection testing has relatively high rates of false-positive results and low positive predictive values. A prospective cohort study of 6662 adults 50 years and older without symptoms of cancer underwent multicancer early detection testing using blood-based cfDNA analysis. Of 6621 participants with analyzable results, cancer signal was detected in 92 participants (1.4%). There was a 38% true-positive rate and a 62% false-positive rate using this general screening strategy.⁵⁴ However, it is possible that cfDNA and ctDNA could improve cancer screening when used as an adjunct diagnostic test and could improve disease-specific screening methods. For example, in average-risk patients eligible for colon cancer screening, a ctDNA test for detection of CRC had an 83.1% sensitivity (95% CI, 72.2%-90.3%) and 89.6% specificity (95% CI, 88.8%-90.3%), demonstrating that this blood-based test may serve as a useful adjunct test for CRC screening in addition to colonoscopy and fecal immunohistochemical testing.⁵⁵

Testing for ctDNA is associated with some challenges. First, testing can be costly to the patient and health care system. Cost of ctDNA testing varies widely according to the type of assay performed. For example, a large genomic panel costs more than an analysis for a specific variant. Costs may also vary according to setting; for example, minimal residual disease testing vs identifying variants for treatment selection. Many personalized minimal residual disease tests can cost between \$3000 and \$5000 or more. When performed every 3 months for surveillance, costs are greater than diagnostic imaging studies for ongoing surveillance. Second, ctDNA test completion may be delayed if there is insufficient capacity to promptly analyze a large number of ctDNA tests. These testing delays may increase anxiety and distress for patients and may delay treatment initiation.

Despite these challenges, ctDNA testing has the potential to improve patient care and long-term outcomes. ctDNA clearance from the blood, which occurs when a patient initially has a positive minimal residual disease test result but later has a negative result, represents a potential end point for future prospective clinical trials. Patients who do not clear their ctDNA on surveillance or minimal residual disease testing could be offered additional treatment or entry into a clinical trial to improve long-term outcomes given the poorer DFS observed in patients with persistently positive ctDNA results.

Limitations

This review had limitations. First, some aspects of liquid biopsies were not discussed. Second, the quality of included studies was not evaluated. Third, some relevant articles may have been missed.

Conclusions

ctDNA, consisting of small DNA fragments from cancer cells that can be analyzed in human blood, can help clinicians monitor cancer progression, detect the presence of minimal residual cancer, and identify genetic variants that may help guide treatment decisions. Use of ctDNA may help select best treatment and timing of therapy for a patient with cancer, but optimal clinical applications remain unclear.

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Submissions: We encourage authors to submit papers for consideration as a Review. Please contact Kristin Walter, MD, at kristin.walter@jamanetwork.org.

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